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Hopping conduction in GeSiSn alloys: Impact of structural disorder on electrical transport

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ABSTRACT

Temperature-dependent charge transport in epitaxial GeSiSn films with varying thicknesses and Si and Sn content, grown on Ge/Si(001) substrates, was investigated using admittance spectroscopy. The conductance exhibits a crossover from Mott variable-range hopping at low temperatures to thermally activated conduction at higher temperatures. Below 150 K, carriers move via localized states in the band tails, with hopping parameters governed by the density and spatial distribution of disorder-induced states. Increasing nominal Sn concentration and thickness enhances structural disorder and Sn segregation, leading to a higher density of states and reduced hopping length. Scanning capacitance microscopy reveals variations in charge-carrier concentration in high-Sn films, indicating the presence of coexisting p-type and n-type regions at the microscale, consistent with compositional fluctuations, Sn segregation, and microstrain. These results demonstrate that transport in GeSiSn alloys is primarily dominated by disorder-assisted hopping at low temperatures, establishing a quantitative link between microscopic disorder and macroscopic electrical response in metastable group-IV semiconductors.

I. INTRODUCTION

Group-IV semiconductor alloys have garnered increasing attention due to their compatibility with existing silicon-based technologies and their potential for novel optoelectronic applications.[1–4] Among these, GeSiSn and GeSn alloys have emerged as a promising material system due to their tunable bandgap, CMOS compatibility, and the ability to engineer electronic properties through compositional and strain variations.[5–7] The possibility of achieving a direct bandgap at sufficiently high-Sn concentrations makes GeSiSn alloys excellent candidates for monolithic on-chip Si photonics operating in the near-infrared and short-wave infrared spectral ranges.[8] In this regard, it is crucial to grow alloys of high crystalline quality and to understand the unavoidable generation of structural imperfections with increasing Sn content. Indeed, the electronic and optical properties of SiGeSn alloys depend strongly on their crystal structure and atomic arrangement, where strain, alloy composition, and defect formation play crucial roles.

Despite the optical and structural characteristics of GeSiSn having been extensively investigated,[4,9,10] electrical transport in GeSn and, especially, in ternary GeSiSn remains comparatively underexplored. In GeSn alloys, the electrical properties are strongly influenced by their microstructure and crystalline defects, including dislocations and point defects that act as sources of shallow and deep traps within the bandgap.[11,12] In addition to the Ge-related traps, GeSiSn alloys may contain additional trap states associated with the incorporation of Si and Sn atoms, whose local bonding configurations and strain fields can further modify the potential landscape, carrier localization, and hopping pathways.

Such complex defect- and strain-induced potential fluctuations are expected to result in the formation of localized states near the Fermi level. In this case, carrier transport often deviates from band-like conduction, instead following a hopping mechanism, especially when localized states dominate.[13,14] Such complex defect- and strain-induced potential fluctuations are expected to result in the formation of localized states near the Fermi level. In this case, carrier transport often deviates from band-like conduction, instead following a hopping mechanism, especially when localized states dominate.[13,14] Structural disorder in group-IV alloys, including point defects, compositional inhomogeneity, strain fields, and short-range ordering (SRO),[15–19] may further enhance carrier localization through nanoscale fluctuations of the band edges. In GeSiSn alloys, the incorporation of both Si and Sn is expected to increase the complexity of these fluctuations due to differences in atomic size and bonding, potentially promoting local ordering and segregation effects that can affect disorder-assisted transport. When these localized states form a percolating network near the Fermi level, charge carriers move primarily through phonon-assisted hopping between localized sites. Depending on the density of states and the strength of Coulomb interactions, the temperature dependence of conductivity follows either the Mott variable-range hopping (VRH) or the Efros–Shklovskii (ES) model, each exhibiting characteristic signatures in both dc and ac (admittance).[20–22] In this context, an open question for GeSiSn alloys is how alloying with Si and Sn modifies the density and distribution of localized states, and consequently how charge transport evolves across different temperature and frequency regimes.

In this work, we use admittance measurements to investigate the influence of GeSiSn composition on transport properties that are crucial for optoelectronic applications. Therefore, we have investigated the hopping conduction behavior in GeSiSn alloys by analyzing the temperature- and frequency-dependence of the complex conductivity to identify the dominant transport mechanisms. Interpreting the data within the VRH model, we extract characteristic parameters that relate charge transport to structural disorder, phase separation, and defect landscapes in these alloys. Our findings provide new insights into the transport physics of GeSiSn and demonstrate how nanoscale inhomogeneity governs low-temperature conduction in coherently strained GeSiSn, offering practical guidelines for tailoring electronic properties in future device implementations.

II. EXPERIMENT DETAILS

Epitaxial $Si_xGe_{1-x-y}Sn_y$ layers were grown on Ge/Si(100) substrates using a reduced-pressure chemical vapor deposition (RP-CVD) system (ASM Epsilon 2000 reactor). Before loading, Si substrates were chemically cleaned using a diluted HF:H_2O (1:20) solution for 1 min to remove the native oxide, followed by nitrogen blow-drying. The substrates were subsequently degassed at 250 °C for over 2 h inside the reactor. Final oxide removal was achieved by high-temperature annealing at approximately 900 °C in a hydrogen ambient before growth. A Ge buffer, with a total thickness of approximately 650 nm, was grown *in situ* before the deposition of the GeSiSn layer using 10% GeH_4 diluted in purified H_2 as the carrier gas.

The buffer was grown in two stages on a boron-doped Si (001) substrate with a dopant concentration of 10^{15} cm^{-3}. First, an initial 150 nm Ge seed layer was deposited at a substrate temperature below 400 °C. This was followed by a temperature ramp-up to 600 °C for the growth of an additional ∼500 nm Ge layer. These two steps were followed by an *in situ* anneal at temperatures exceeding 800 °C to improve crystal quality and surface morphology. After annealing, the substrate was cooled to 400 °C to initiate the growth of the GeSiSn films. Gas flow ratios of SiH_4/GeH_4 were varied between 0.15 and 0.6, while the $SnCl_4/GeH_4$ mass flow ratio was fixed at 0.01.

Two thickness series of epitaxial GeSiSn films were grown to probe thickness-dependent strain relaxation and structural evolution. Series A used a nominal composition of 7.3 at. % Si and 5.5 at. % Sn, while Series B used 12 at. % Si and 9 at. % Sn. Within each series, three samples were grown with active-layer thicknesses of from 50 to 200 nm. The sample parameters are summarized in Table I. A reference Ge/Si sample without a GeSiSn film was also studied for comparison.

For electrical measurements, the samples were cut into 5×8 mm^2 pieces. Subsequently, Ga–Zn eutectic contacts were deposited onto the GeSiSn layers at 170 °C. The contacts were 5×1 mm and separated by 6 mm. The experimental *I–V* curves at an applied voltage below 1 V were found to be linear in the range from 80 to 300 K, demonstrating Ohmic behavior. Admittance spectroscopy measurements were performed in the temperature range of 80–300 K, using an LCR Measuring Bridge ST2829C (SourceTronic) at a modulation voltage of 20 mV and in the frequency range from 1 kHz to 1 MHz.

X-ray diffraction measurements were performed using the Rigaku SmartLab Multipurpose X-ray Diffraction System equipped with a 3.0 kW Cu x-ray tube, four-bounce Ge(220) monochromator,

TABLE I. Film parameters for GeSiSn/Ge/Si(001) heterostructures.

Sample name	Composition of Sn (%)	Nominal composition of Si (%)	Nominal | Estimated film thickness (nm)	Strain (004) out-of-plane (%)	RC FWHM (004) (arc sec)	RMS roughness (nm)
A1	5.35	7.3	50 | 56	0.22	284.9	1.8
A2	5.26	7.3	150 | 149	0.20	193.6	2.8
A3	5.31	7.3	200 | 199	0.16	178.0	5.5
B1[a]	8.22/4.53	12	100 | 20/80	0.28/0.09	576.1	13.2
B2[a]	7.80/4.52	12	150 | 30/120	0.23/0.05	563.4	15.5
B3[a]	7.77/4.58	12	200 | 40/160	0.21/0.04	583.4	23.9

[a]The films revealed two GeSiSn layers with different compositions.

10.0 mm height limiting slit, four-bounce Ge(220) channel-cut monochromator, and the HyPix-3000 pixel detector. The reciprocal-space maps (RSMs) were acquired in the vicinity of the asymmetric (224) reflection to quantify the in-plane lattice parameter and the degree of relaxation. Complementary symmetric 2θ–ω scans and ω-rocking curves around the (004) reflection were used to extract the out-of-plane lattice parameter/strain and the RC FWHM.

Surface topography and lateral carrier inhomogeneity were characterized with a Dimension 3000 NanoScope IIIa scanning probe microscope (Bruker AXS, formerly Digital Instruments) employing a contact-mode atomic force microscopy (AFM)/scanning capacitance microscopy (SCM) measurement protocol. Atomic force microscopy (AFM) was used to acquire surface topography. At the same time, scanning capacitance microscopy (SCM) data were collected from the identical points. The measurements were carried out using solid Pt tips (nominal radius ∼10.00 nm). Amplitude and phase of the capacitance-modulation signal were acquired with an ac bias of ∼1.00 V at 100.00 kHz applied between a surface contact on the GeSiSn film and the SCM tip (no dc offset unless specified). All AFM/SCM measurements were conducted under ambient laboratory conditions (∼23.00 °C, relative humidity ∼40%).

III. RESULTS

A. Material characterization

To study the crystalline quality of the epitaxial layer and its potential influence on transport properties, we employed XRD and atomic force microscopy (AFM). Figures 1(a) and 1(b) show the AFM surface topography images of representative GeSiSn surfaces from samples A2 and B2, illustrating the evolution of surface morphology with varying Si and Sn content. Series A (lower Sn; sample set A1–A3) exhibits RMS roughnesses increasing from 1.8 to 5.5 as the GeSiSn film increases in thickness from 50 to 200 nm. The surface remains laterally homogeneous and is decorated by weak, discontinuous ridge-like features aligned with the orthogonal ⟨110⟩ directions (see Fig. S1 in the supplementary material). Such morphology may reflect the onset of strain relaxation; however, the exact origin of these ridge-like features remains unclear. Unlike the well-developed cross-hatch patterns typically associated with dislocation-mediated relaxation,[23,24] the observed short segments aligned along ⟨110⟩ directions are more likely related to localized defect structures, such as stacking faults, or other strain-relief mechanisms characteristic of low-temperature growth. The topography is, therefore, consistent with a kinetically limited regime in Sn-containing group-IV alloys grown at ≤400 °C, where defect formation and strain accommodation occur locally rather than through extended dislocation glide, resulting in short, discontinuous features instead of long continuous ridges typical of high-temperature annealed samples.[25,26] In contrast, Series B (higher Sn; samples B1–B3) transitions to a markedly rougher surface, with RMS roughnesses from 13.20 to 23.90 nm. This is consistent with Sn segregation and phase separation during growth, as evidenced by the presence of droplet-like surface features observed in AFM topography images of all GeSiSn films from series B (Fig. S1 in the supplementary material). Nanometer-scale trenches attributable to droplet-mediated surface migration are observed, which also agrees well with the prior reports.[27]

Figures 1(c) and 1(d) present reciprocal-space maps (RSMs) around the (224) reflection for the GeSiSn/Ge/Si heterostructures A2 and B2 plotted in lattice-parameter coordinates ($a_{\parallel}$, $a_{\perp}$). The Si substrate, relaxed Ge buffer, and GeSiSn layer lattice points are observed in each map. In a representative Series A map, the GeSiSn lattice point lies directly above the Ge-buffer position at the same $a_{\parallel}$, confirming pseudomorphic in-plane growth; the

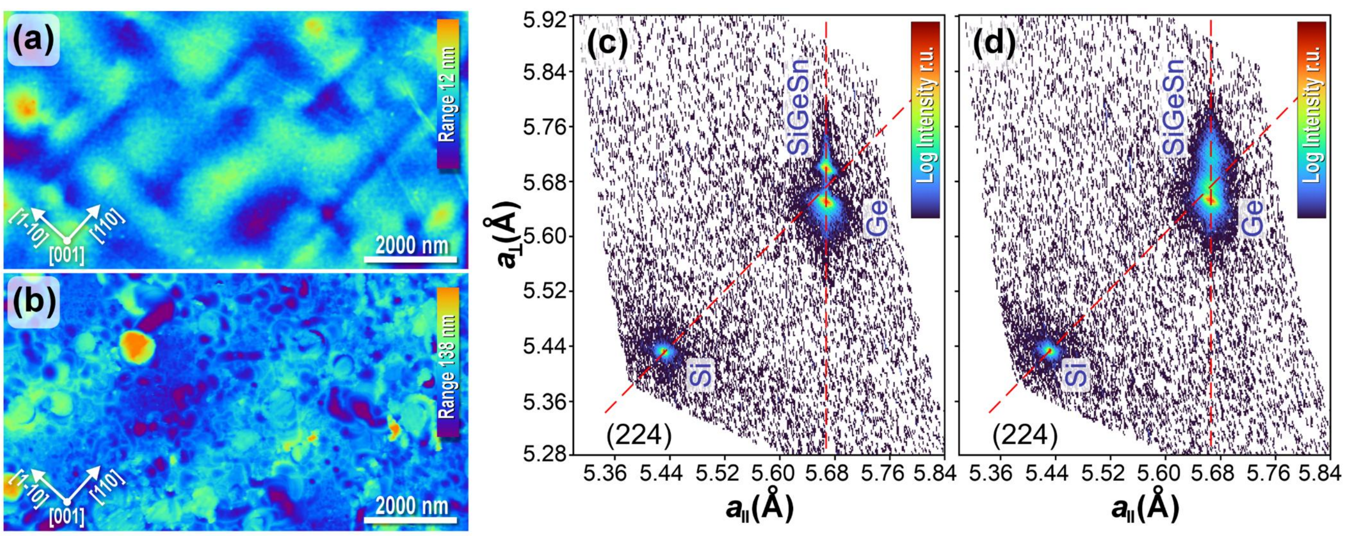

FIG. 1. (a) and (b) AFM topography of $Si_xGe_{1-x-y}Sn_y$/Ge/Si structures A2 and B2; (c) and (d) HRXRD RSMs around the asymmetric (224) reflection. Red dashed lines indicate the full-relaxation line (diagonal) and the in-plane lattice parameter of Ge (vertical). Panel mapping: A2 → (a,c), B2 → (b,d). Thickness of both films: 150.00 nm.

upward shift in $a_\perp$ reflects the expected tetragonal distortion. By contrast, in Series B, the GeSiSn lattice point is broadened and exhibits a clear vertical doublet along $a_\perp$ [see Fig. 1(d)], indicating a bilayer or phase-separated structure with different Sn contents and residual strains, consistent with the AFM topography. However, the film remains pseudomorphic, as evidenced by the vertical alignment of both diffraction features with the Ge layer. According to the RSMs, the in-plane strain $\varepsilon_\parallel$ varies from −0.39% to −0.32% for Series A films and from −0.64% to −0.70% for Series B films. Composition, strain, and thickness were extracted from the analysis of asymmetric (224) RSMs, complemented by symmetric 2θ–ω scans and ω-rocking curves around the (004) reflection. The resulting values are summarized in Table I. The Si content was fixed at its nominal value in the modeling, assuming stable incorporation during CVD growth. The complete set of AFM, RSM, and 2θ–ω scans and ω-rocking curves is provided in the supplementary material.

The estimated Sn content for Series A was close to the nominal target of 5.5% (see Table I). Increasing thickness from 50 to 200 nm decreases the out-of-plane strain from about 0.22% to 0.16%, while the FWHM of the (004) rocking curve narrows from ∼285 to ∼178 arc sec. The strain relaxation is consistent with the surface undulations observed by AFM, while the layers remain pseudomorphic in-plane with respect to the Ge buffer, as confirmed by the identical $a_\parallel$ values in the RSMs.

In Series B, however, the rocking-curve FWHM systematically increases with thickness. At the same time, the GeSiSn lattice point shows broadening and splitting along $a_\perp$. Joint fits of symmetric 2θ–ω scans containing two peaks associated with the GeSiSn layer (see Fig. S3 in the supplementary material) and the asymmetric RSMs are consistent with a vertically graded structure consisting of a sublayer with Sn content of about 8% and out-of-plane strain $\varepsilon_\perp \approx 0.21\%–0.28\%$ and a sublayer with Sn content of 4.5% that is nearly relaxed with $\varepsilon_\perp \approx 0.04\%–0.09\%$ (see Table I). In such a vertically heterogeneous film, the measured ω-scan integrates contributions from sublayers with different $a_\perp$ and tilt distributions, naturally producing a broader effective FWHM.

Taken together, AFM and HRXRD provide the following structural picture. At moderate Sn (Series A), films grow pseudomorphically with low mosaicity. With increasing film thickness, partial strain relaxation occurs,[11] typically accompanied by an increase in surface roughness. The observed reduction in the rocking-curve FWHM with thickness is attributed to a combination of factors, including increased scattering volume and improved structural coherence away from the Ge/GeSiSn interface.[28] The in-plane strain of the GeSiSn thin films remains weakly thickness dependent from $\varepsilon_\parallel \approx -0.39\%$ to −0.32% across the series. For higher-Sn content (Series B), the growth temperature is sufficient to kinetically enable Sn segregation and vertical content gradient in the metastable GeSiSn alloy. Effective in-plane strains vary from $\varepsilon_\parallel \approx -0.39\%$ to −0.70% across the series; within a given bilayer, the Sn-richer lower sublayer retains the higher compressive $\varepsilon_\parallel$, whereas the Sn-poorer top sublayer is close to relaxed (as also reflected by the small $\varepsilon_\perp$). This ordering pathway, in particular, coherent, laterally smooth layers at lower Sn vs vertically compositionally varying, rougher layers at higher Sn, provides a physically self-consistent framework that reconciles the observed FWHM trends, AFM roughening, and RSM doublets, and points to a correspondingly more disordered electronic landscape in the investigated high-Sn samples.

B. Frequency-dependent conductivity

Figure 2(a) shows the frequency dependence of the real part of the admittance of the samples with GeSiSn films on the Ge/Si substrate and the reference sample without GeSiSn, measured at 85 K. The conduction of the samples with the GeSiSn layer was significantly higher than that of the reference sample, which is attributed to the dominant contribution of the GeSiSn layer to the overall conductivity. The Nyquist plot of the imaginary part of the impedance (Z_2) vs the real part (Z_1), as shown in Fig. 2(b), exhibits a depressed single semicircular arc, indicating that the electrical processes in the GeSn films can be modeled using an equivalent circuit consisting of a resistor in parallel with a constant phase

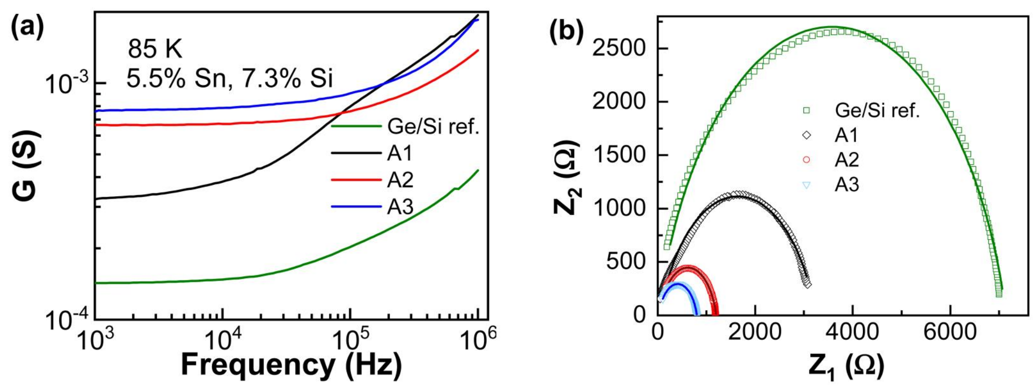

FIG. 2. (a) Real part spectrum of the complex conductivity measured at 85 K; (b) Nyquist plots (Z_2 vs Z_1) for samples with GeSiSn films on Ge/Si substrates and for the reference sample without the GeSiSn layer.

element (R-CPE), indicating that the electrical response of the GeSiSn films is governed by a single dominant relaxation process with non-ideal capacitive behavior. A slight asymmetry of the arcs, particularly noticeable for A1 and the Ge/Si reference, is attributed to the distribution of relaxation times, as well as to minor contributions from contact/interface effects and residual parasitic elements at low frequencies. This deviation is weak and unlikely to affect the validity of the equivalent-circuit modeling or the extracted transport parameters.

The conductivity data were modeled using an equivalent circuit consisting of a resistor (R) connected in parallel with a constant phase element (CPE). The admittance of the parallel combination of resistor R and CPE element is

$$Y(\omega) = \frac{1}{R} + Q(j\omega)^s. \tag{1}$$

Thus, impedance is given by

$$Z(\omega) = \frac{1}{\frac{1}{R} + Q(j\omega)^s}. \tag{2}$$

Therefore, the real (frequency-dependent) conductivity $\sigma'(\omega)$ is the real part of the admittance,

$$\sigma'(\omega) = \frac{1}{R} + Q\omega^s, \tag{3}$$

where R is the resistance reflecting DC conduction, Q is the CPE parameter representing non-ideal capacitive behavior, s is the CPE exponent (between 0 and 1), indicating the deviation from ideal capacitive behavior (see Figs. S4 and S5 in the supplementary material).

This model assumes that the GeSiSn thin films exhibit two main conduction mechanisms. The frequency-independent conduction is associated with direct conduction pathways through extended states. In contrast, frequency-dependent conduction (dispersive conduction) characterizes the dispersive electrical response, leading to non-ideal capacitive effects due to hopping, trapping, or interface phenomena.

Admittance measurements were performed in the temperature range of 85–300 K, with the real part of the admittance fitted using Eq. (3) to investigate the transport mechanism along the epitaxial layer. The temperature dependence of extracted DC conductance, $G_{DC} = 1/R$, for GeSiSn epitaxial layers with different thicknesses is shown in Fig. 3. The conductance of the Ge/Si reference layer was much lower at low temperatures, indicating the dominance of conduction paths through GeSiSn thin films and GeSiSn/Ge interface states [see Fig. S6(a) in the supplementary material]. The temperature-dependent DC conductance of GeSiSn epitaxial layers reveals insights into the underlying transport mechanisms. At higher temperatures (above ∼200 K), the conductance is influenced by parallel conduction paths through the GeSiSn layer, the Ge buffer, the Si substrate, and interfaces, as well as temperature-dependent carrier redistribution within the heterostructure. In this temperature range, carrier mobility decreases due to enhanced phonon scattering,[29] which tends to reduce the conductance with increasing temperature [see Fig. 3(b)]. As the temperature decreases, particularly below 120 K, a deviation from Arrhenius-type behavior is observed, indicating the onset of hopping conduction. To test the Efros–Shklovskii variable-range hopping model, G_{dc} was plotted on a logarithmic scale as a function of $T^{-1/2}$. The resulting dependence is non-linear, indicating that the ES model does not provide an adequate description of charge transport in the studied temperature range (see Fig. 7S in the supplementary material). In contrast, the low-temperature conductivity exhibits a linear dependence in a $\ln(G_{dc})$ vs $T^{-1/4}$ representation, consistent with three-dimensional Mott variable-range hopping.

C. Variable-range hopping in GeSnSi alloys

In the case of three dimensions, the conductivity in disordered materials follows the Mott law:[30]

$$\sigma = \sigma_0 \exp\left(-(T_0/T)^{1/4}\right), \tag{4}$$

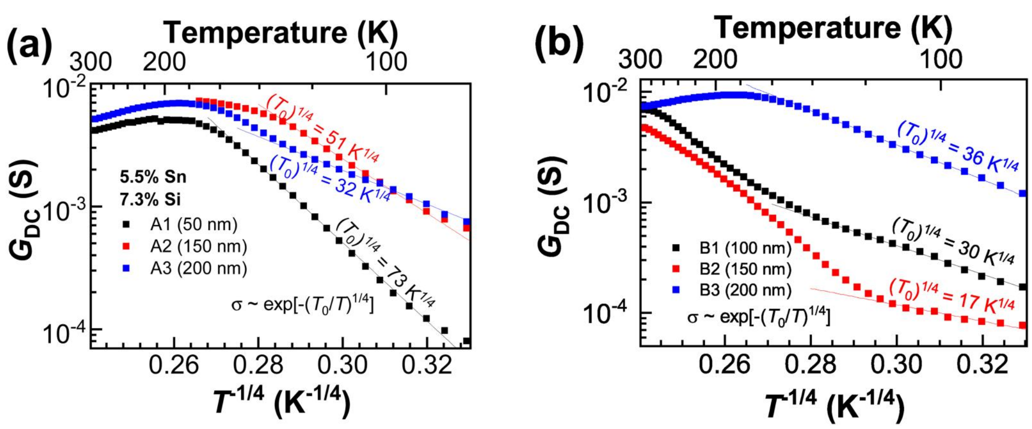

FIG. 3. Temperature-dependent DC conductance plotted as a function of $T^{-1/4}$ for GeSiSn epitaxial layers with different thicknesses (50, 150, and 200 nm) containing 5.5% Sn and 7.3% Si (a) and 9% Sn and 12% Si (b).

TABLE II. Density of states and hopping parameters evaluated at 85 K, calculated using Eqs. (5)–(7).

Sample	$(T_0)^{1/4}$ ($K^{1/4}$)	$N(E_F)$ ($\times 10^{19}$ eV^{-1} cm^{-3})	R_{hop} (nm) at 85 K	W (eV)
Seria A				
A1	73.4	1.64	19.4	3.48
A2	51.0	7.02	13.5	0.814
A3	31.6	47.5	8.4	0.120
Seria B				
B1	29.8	60.1	7.9	0.095
B2	17.1	561	4.52	0.010
B3	35.8	28.9	9.49	0.198

where σ_0 and T_0 are material-dependent parameters. The pre-exponential factor σ_0 is typically weakly dependent on temperature. The characteristic Mott temperature T_0 is related to the density of localized states $N(E_F)$ near the Fermi level and the localization radius of the carrier wave function ξ,

$$T_0 = \frac{\beta}{k_B N(E_F)\xi^3}, \tag{5}$$

where $\beta = 21$[20] and k_B is the Boltzmann constant.

The expression for the hopping length, R_{hop}, and the activation energy for hopping, W, is given by

$$R_{hop} = \xi\left(\frac{T_0}{T}\right)^{1/4}, \tag{6}$$

$$W = \frac{1}{4}k_B T_0\left(\frac{T_0}{T}\right)^{1/4}. \tag{7}$$

The VRH parameters, such as values of σ_0 and T_0, and the calculated $N(E_F)$, R_{hop}, and W are presented in Table II. For the low-Sn series (samples A1–A3), $N(E_F)$ increases from 1.64×10^{19} to 4.75×10^{20} $eV^{-1}\,cm^{-3}$ with increasing thickness, while the corresponding hopping length decreases from 19.4 to 8.4 nm at 85 K. This trend indicates that as the density of localized states rises, electronic transport is facilitated by shorter hops between localized sites. In the higher-Sn series (samples B1–B3), some temperature dependencies of conductance reveal two sections with different slopes [Fig. 3(b)], and only low-temperature ones with gentler slopes may be related to hopping behavior. At the same time, the more rapid increase at higher temperatures may be attributed to the thermal activation of the carriers and additional parallel conduction pathways associated with metallic Sn-rich droplets, as indicated by the noticeably higher conductance of this sample already at low temperatures. The estimated hopping distances range from 4.5 to 9.5 nm, accompanied by comparatively high densities of localized states on the order of 10^{20}–10^{21} eV^{-1} cm^{-3}. The thickness dependence for this series is non-monotonic: the shortest hopping distance, $R_{hop} = 4.52$ nm, is observed for sample B2 (150 nm), whereas in sample B3 (200 nm) R_{hop} increases to 9.49 nm, a value comparable to sample B2. Sample B2 exhibits the highest density of states, which agrees with the RSM and AFM observations that indicate cluster formation and increased structural disorder. At the same time, the thickest film (B3) appears to accumulate additional structural nonuniformity associated with the presence of segregated Sn-rich droplets, which broadens the distribution of localized states and simultaneously reduces their effective spatial overlap. The evolution of the density of states with increasing Sn content or film thickness reflects a competition between generation of defect states and Sn segregation: while Sn incorporation tends to increase $N(E_F)$, Sn segregation redistributes localized states and may reduce their effective density near the Fermi level. This highlights the combined role of disorder, segregation, and alloy composition in defining low-temperature charge transport.

D. Scanning capacitance microscopy

To assess the lateral homogeneity of the electrical response, we carried out SCM mapping of the GeSiSn films under the conditions described in Sec. II. The measurements were performed through the native surface oxide and adventitious adsorbates, which act as a thin dielectric spacer between the conductive tip and the semiconductor. In this configuration, the dC/dV phase provides a robust qualitative indicator of the local majority-carrier type (our sign convention: positive dC/dV phase leads to prevailing p-type response).[30]

The insets in Fig. 4 show SCM phase images for the surfaces of the 150-nm thick GeSiSn thin films (samples A2 and B2). The top SCM image (a) reveals only a positive phase shift that reflects a p-type response, which is expected for grown GeSn without intentional doping, as having vacancies and dislocations.[11] Meanwhile, the sample of series B (b) shows pronounced phase contrast with phase-inverted domains: regions of positive phase (local p-type) coexist with patches of negative dC/dV phase shift corresponding to n-type behavior with locally enhanced electron concentration. These n-type patches appear with typical lateral sizes of $\sim$0.50–$\sim$2.00 μm.[30] The broad phase histogram reflects strong lateral inhomogeneity of the electronic landscape, which is correlated with the presence of Sn droplet-like surface features observed in AFM topography and with the broadening and splitting of diffraction features in HRXRD/RSM, indicative of vertical compositional inhomogeneity. The appearance of mixed-polarity domains, i.e., fluctuations in charge-carrier concentration, is consistent with local band-bending fluctuations and trap/disorder

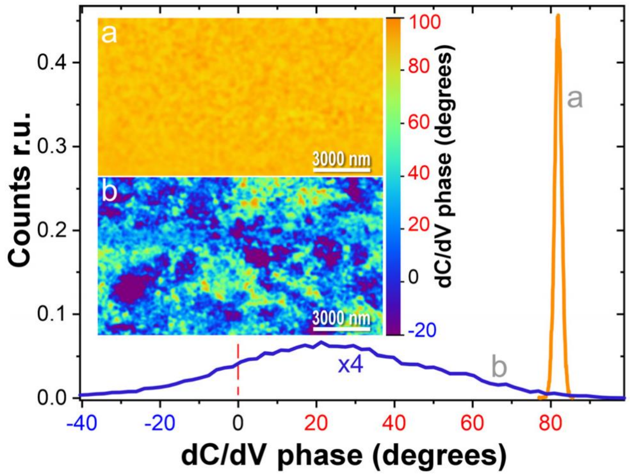

FIG. 4. Histograms of the SCM dC/dV phase with the corresponding phase maps shown as insets for (a) sample A2 and (b) sample B2. The phase maps are displayed using the same color scale. The sign of the dC/dV phase is a qualitative indicator of the majority-carrier type; a positive phase denotes a prevailing *p*-type response (the vertical dashed line marks 0°). The histogram (b) is multiplied by 4 for better visualization.

distributions caused by Sn segregation and other local inhomogeneities (strain fields, defects, etc.). Overall, the SCM results corroborate the structural analysis: Series A films are laterally uniform and *p*-type within our resolution, whereas Series B films exhibit mesoscale electronic phase separation that mirrors their structural/compositional heterogeneity.

IV. DISCUSSION

The results suggest that charge carriers in the studied GeSiSn films predominantly percolate via Mott VRH at low temperatures through disorder-induced localized states, whereas at higher temperatures, thermally activated conduction becomes dominant. The observed crossover between these two regimes reflects the balance between defect-related band-tail states and extended states, with the relative contribution governed by film thickness, Sn content, and the associated degree of structural disorder.

At temperatures above ∼120 K, the conductivity of the GeSiSn alloys deviates from the Mott VRH law due to the observation of a thermally activated contribution. In this regime, charge carriers gain sufficient energy to be excited from localized band-tail states or defect levels into extended states. It leads to an Arrhenius-type temperature dependence of conductivity ∼exp $[-E/(k_BT)]$ with activation energies E of the order of several tens of meV. The plot of the temperature dependency of conductance in the respective scales (see Fig. S6 in the supplementary material) does not allow estimation of the activation energy precisely, as the low-temperature part of the slope is distorted, perhaps due to the presence of a hopping mechanism of conductivity and, maybe, carrier activation from deep defect levels with different depths. The activation-energy values of tens of meV extracted from the slopes at higher temperatures correspond to the depths of the acceptor- and, partly, donor-like defects responsible for the thermally activated conduction.[31]

The analysis of VRH parameters given by Eqs. (5)–(7) demonstrates that the density of localized states strongly governs charge transport in GeSiSn alloys at low temperatures (below 120 K). In the low-Sn series, $N(E_F)$ increases with thickness while the rocking-curve FWHM decreases, indicating that improved crystalline mosaicity with thickness does not necessarily imply a reduced density of electrically active localized states; the latter can increase due to the evolution of misfit dislocations during strain relaxation and associated local compositional and strain fluctuations, which contribute to the formation of band-tail states responsible for hopping conduction. The positive dC/dV phase shift and minor variations observed for samples A1–A3 indicate p-type behavior with a uniform surface potential distribution. By contrast, the higher-Sn series (B1–B3) exhibits larger dC/dV phase dispersion and amplitude contrast (including positive and negative phase shifts), consistent with enhanced microstrain and alloy clustering that produce spatially varying carrier density; this correlates with higher $N(E_F)$ and shorter hopping lengths extracted from Eqs. (5)–(7). In the higher-Sn series, the hopping distances are consistently shorter (4.5–9.5 nm) and are associated with even higher densities of localized states (∼10^{20}–10^{21} cm^{-3}), indicating that alloy clustering and microstrain generate dense localized states near the Fermi level that strongly participate in low-temperature transport.

From a general perspective, short-range ordering, point defects, and compositional fluctuations in GeSiSn alloys are expected to give rise to a network of localized states within the mobility gap. At low temperatures, carriers percolate via VRH across band-tail states, with the hopping length defined by the density and spatial correlation of these localized states. The extracted VRH parameters are consistent with a disorder-assisted hopping regime in GeSiSn, whose characteristics correlate with film thickness and alloy composition, and are further influenced by Sn clustering and associated structural inhomogeneity. This interpretation aligns with the nanoscale inhomogeneities observed in SCM maps (see Fig. 4). In particular, SCM images reveal fluctuations in carrier concentration originating from alloy clustering and surface defect generation. Such electronic inhomogeneities are consistent with the shorter hopping distances and higher $N(E_F)$ values extracted for Sn-rich films (e.g., samples B1–B3). This correlation suggests that structural and compositional disorder, even at a thickness of 100 nm, provides additional carrier transport pathways. For seria B, increasing Sn + Si content does not simply lead to more tail states near the Fermi level. Instead, the changes in $N(E_F)$ were influenced by Sn segregation, which induces a micro-scale compositional and electrostatic potential fluctuations. These fluctuations can lead to spatial variations in near-surface band bending and carrier re-distribution, resulting in contrast in SCM measurements that may appear as regions with p-type and n-type response (see Fig. 4). We note that the observed n-type-like response may arise from electron accumulation near Sn-rich regions or from the presence of metallic Sn droplets affecting the local capacitance signal. This interpretation is supported by XRD, which shows that high-Sn alloys exhibit more substantial peak broadening (indicative of microstrain and phase separation) yet remain pseudomorphic.

The interplay between structural disorder, nano- and micro-scale compositional fluctuations, and strain fields (rather than macroscopic relaxation) sets the density and spatial distribution of localized states in the investigated GeSiSn films. These states, in turn, govern the crossover from Mott VRH at cryogenic temperatures to thermally activated conduction at higher temperatures. The results underscore the central role of disorder-assisted transport in GeSiSn, particularly at high-Sn incorporation, where defect formation and strain-driven instabilities are most pronounced.

V. CONCLUSION

Impedance spectroscopy of GeSiSn alloys reveals that low-temperature charge transport is governed by Mott variable-range hopping through disorder-induced localized states, whereas at higher temperatures, conduction transitions to a thermally activated regime. The extracted VRH parameters indicate that increasing film thickness and higher nominal Sn and Si content, accompanied by partial Sn segregation, enhance the density of localized states and reduce hopping distances, reflecting the combined roles of structural defects, alloy clustering, and microstrain in carrier localization. SCM mapping and complementary structural analyses support this interpretation, revealing nanoscale inhomogeneities in carrier density correlated with alloy clustering and microstrain. Sn-rich samples exhibit a higher degree of defect-related disorder, an increased density of localized states near the Fermi level, and the presence of coexisting p-type and n-type regions at the microscale, as revealed by SCM. Overall, the results suggest that charge transport in GeSiSn alloys may be associated with disorder-assisted hopping through a network of localized states arising from short-range ordering, strain fields, and compositional fluctuations. The observed VRH conduction and its dependence on Sn content and thickness highlight the role of structural disorder in carrier dynamics in metastable group-IV alloys. These insights provide a microscopic basis for optimizing GeSiSn-based infrared and electronic devices, where control of strain, defect density, and alloy uniformity is crucial for tailoring electrical and optical performance.

SUPPLEMENTARY MATERIAL

The supplementary material includes results of AFM analysis: Topography maps ($10 \times 10\,\mu m^2$) of the GeSiSn films with indicated RMS roughness values; XRD analysis: Reciprocal-space maps (224) of GeSiSn/Ge/Si(001) heterostructures demonstrating pseudomorphic GeSiSn layers on Ge. High-resolution 2θ–ω and ω-scan data illustrate the dependence of structural disorder on film thickness and Sn composition. The relaxed lattice parameter and strain components were evaluated using Vegard's law and biaxial strain relations; "Impedance analysis: Frequency-dependent conductance and Jonscher power-law fitting" presents the spectra of the real part of impedance, shows temperature-dependent variations of the exponent s, and depicts Arrhenius plots of DC conductance for activation-energy estimation.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Serhiy Kondratenko: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **Oleksandr Datsenko:** Formal analysis (equal); Investigation (equal); Visualization (equal). **Hryhorii Stanchu:** Investigation (equal). **Andrii Pocherpailo:** Investigation (equal); Visualization (equal). **Petro M. Lytvyn:** Conceptualization (equal); Visualization (equal); Writing – review & editing (equal). **Morgan E. Ware:** Investigation (equal); Writing – review & editing (equal). **Serhii Maliuta:** Investigation (equal). **Shui-Qing (Fisher) Yu:** Formal analysis (equal); Methodology (equal). **Yuriy I. Mazur:** Investigation (equal); Methodology (equal); Writing – original draft (equal). **Gregory J. Salamo:** Conceptualization (equal); Formal analysis (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its supplementary material.

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